AMORTIZED EXPLOITABILITY CERTIFICATES FOR LEARNED RESPONSE EQUILIBRIA

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ABSTRACT

We develop an amortized exploitability certificate for fixed points of learned response maps in games with compact convex product action sets and costs convex in each player's own action. We certify the exploitability of arbitrary strategy profiles with playerwise envelopes without requiring contraction, monotonicity, or uniqueness of equilibrium. Its two game-dependent quantities are each player's own-gradient norm and the consistency error between the learned response operator and the pseudo-gradient. Both are enveloped once by cost-oracle probes on a finite η-covering of the certified region Ω. Subsequent profiles require only a forward pass. Amortization is a feasibility requirement because at profiles where players' projections are inactive, single probes cannot improve the model-free bound. The price of amortization is that coverings at fixed accuracy grow exponentially with the dimension of Ω. We evaluate our results across two product-set games and seven response families. We find that small-gain contraction certification is neither necessary nor sufficient for low exploitability in the models tested; and on a non-convex game the natural residual vanishes analytically at a quarter-exploitability profile, so that playerwise convexity becomes a strict requirement.

1 INTRODUCTION

Decision-making in multi-agent settings can be characterized as *non-cooperative games*, where heterogeneous and selfish entities, i.e. *agents or players*, act with independent decisions to optimize their own objective, irrespective of other players' goals. With applications that traverse socio-technical systems (Ekpo et al., 2026), regional science and transportation planning (Facchinei & Pang, 2003), multi-agent/robot path-finding (Zhang et al., 2024; Wang et al., 2024; Ren et al., 2021), and multi-agent reinforcement learning (Zhang et al., 2021), these games are often encumbered with incomplete or conflicting information and multiple conflicting goals. The (generalized) Nash equilibrium (Facchinei & Kanzow, 2010; Pavel & Hong, 2022; Li et al., 2022), is the solution concept used to characterize the strategic interactions and possible outcomes of such games. Modeling the players' interactions and analyzing the resulting equilibria beyond a small number of players proves intractable: the normal-form representation grows exponentially in the number of players, and even computing an approximate equilibrium is PPAD-hard (Rubinstein, 2015). Beyond tractability, for systems where cost functions are unavailable or players are boundedly rational, the Nash solution concept is practically ineffective for characterizing best responses.

This has stimulated the computation of equilibria from players' interaction data rather using structured learning principles (Scarabaggio et al., 2024). These data-driven formulations extend fixed-point theorems that analyze solution existence under convexity (Kakutani, 1941) to response analysis from data. Relaxations such as the first-order Nash equilibrium notion of Scarabaggio et al. (2026) retain the players' objectives while weakening the equilibrium concept, so that each player's action is a fixed point of its own projected gradient, with existence following from algebraic topology rather than from convex analysis. Scarabaggio et al. (2024) replace the objectives, characterizing best responses as a composition of differentiable neural network layers fitted to players' observed actions. Feasibility is enforced by a projection of the network output onto each player's strategy set. Related formulations learn a parametric equilibrium map trained against the Nikaido–Isoda gap instead of regressing on observed play (Bemporad & Tatarenko, 2026a), or take the reacting object to be the data distribution rather than the response (Narang et al., 2023). This paper works in the second setting and certifies

against the first: we measure the residual of the learned response map, which costs one forward pass, and bound through it the projected-gradient residual whose zeros define the first-order notion.

Suppose that $\mathcal{N} := \{1, \dots, N\}$ indexes the players and player i chooses an action x_i from a nonempty compact convex set $X_i \subseteq \mathbb{R}^{n_i}$. Learning-based Nash equilibrium (LBNE) (Scarabaggio et al., 2024) fits a map $f_i(x_{-i})$ predicting player i's response to the other players' actions, x_{-i}. The *behavioral equilibrium* is a profile $x^\star$ for which $x_i^\star = f_i(x_{-i}^\star)$ for every $i \in \mathcal{N}$. Each f_i composes a learned observer, a response head, and a Euclidean projection onto X_i, so that feasibility of the response is enforced by the projection. Given the continuity of each f_i and the convexity and compactness of the product X, Scarabaggio et al. (2024) establish that the joint map $F = (f_1, \dots, f_N)$ admits a fixed point by Brouwer's theorem (Brouwer, 1911). However, convergence of the prescribed Picard–Banach iteration $x^{k+1} = F(x^k)$ is secured only under an aggregate contraction condition $\left(\sum_{i \in \mathcal{N}} \gamma_i^2\right)^{1/2} < 1$ on the playerwise Lipschitz constants γ_i, which excludes the nonexpansive case at which the iteration can fail to converge at all. Contraction hypotheses of this kind are replete across response-learning and decision-dependent formulations, where they buy uniqueness, a linear rate, and a perturbation bound $\varsigma/(1-q)$ under a uniform map error ς (Mignoni, 2025; Narang et al., 2023; Bauschke & Combettes, 2017).

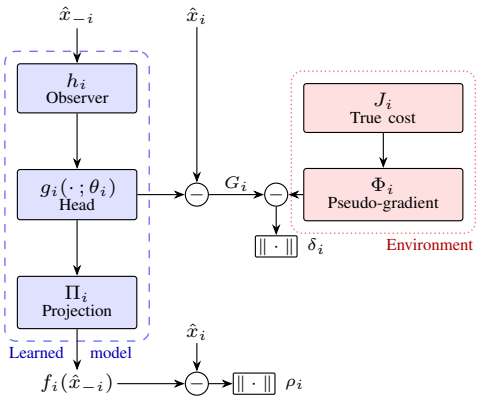


Fig. 1: **A learned response map, and the game it is fitted to.** Both G_i (6) and ρ_i (9) are differences from the same $\hat{x}_i$; δ_i is the consistency error, the one discrepancy that cannot be formed from the model alone.

These guarantees constrain the learned map rather than the game. A contractive F has a unique fixed point of the *learned* game, which need not be an equilibrium of the game that generated the data: the certificate bounds the sensitivity of a fitted network, whereas the quantity of interest is the regret a player would incur at the predicted profile. For arbitrary strategies $\hat{x} \in X$, a constant map $F \equiv \hat{x}$ is nonexpansive and has zero fixed-point residual at $\hat{x}$, while its exploitability is determined by the environment alone and is unconstrained by that residual. Enforcing contraction restricts the hypothesis class where the restriction is least affordable i.e. an interaction whose true best-response map is not contractive admits no contractive model since multi-agent retraining dynamics admit chaotic orbits (Piliouras & Yu, 2023).

While Scarabaggio et al. (2024) introduced the projected observer–head response map and showed that its fixed point solves a variational inequality of the learned game, Narang et al. (2023) obtained uniqueness and a linear rate for performatively stable equilibria under an analogous ratio condition. In a similar vein, Bemporad & Tatarenko (2026a) trained a network against the Nikaido–Isoda gap (Nikaidô & Isoda, 1955) of a surrogate game, while Bemporad & Tatarenko (2026b) instead enforce monotonicity of the learned game by construction. In contrast, Gürkan & Pang (2009) characterized when equilibria of approximating games converge to equilibria of the true game; however, the results are asymptotic with no explicit constant, and Bergerault et al. (2026) proved that the Nash gap, the exploitability of a profile, is *not* determined uniformly over games by a measure-matching error, establishing a divergence between prediction loss and any equilibrium guarantee. Neither approach yields a number that can be evaluated at a given profile *in-silico*: the first certifies the model under a hypothesis that may fail on the interaction of interest, and the second delimits what fitting can achieve without quantifying the shortfall anywhere. A certificate should therefore bound regret in the true game of the prediction model's profile, and should (ideally) be finite when the learned map does not contract.

Write ρ_i for player i's fixed-point residual, δ_i for the consistency error between the learned response operator and the true pseudo-gradient, and $\bar{\delta}_i$ for a bound on δ_i over the certified region (see Fig. 1). Our **contributions** are as follows: (1) We certify the exploitability of every profile $\hat{x} \in \Omega$ with a playerwise envelope, without requiring contraction, monotonicity, or equilibrium uniqueness. (2) We derive a quadratic exploitability bound in the distance to an interior best response, while the general

envelope certificate remains linear in $\rho_i + \bar{\delta}_i$. (3) We show that the residual entry earns its cost only under amortization: $\bar{\delta}_i$ is enveloped once on a certified region using a finite covering, after which each residual costs one forward pass. (4) Across the two product-set games with analytic or numerically verified ground truth and seven response families, the bound holds in all 280 theorem-supported cells. The remaining 140 cells use a coupled feasibility constraint and are reported only as a stress diagnostic. Pooling the certifiable product-set cells with a coupling sweep, a small-gain contraction test admits a fraction 0.09 of the 1000 theorem-supported cells, against a fraction 0.524 whose exploitability is at or below 0.01, so that certification and equilibrium quality come apart.

The rest of this paper is organized as follows: Section 2 formulates the game and its learned response map, Section 3 states the certificate, Section 4 gives the envelope and its cost, and Section 5 reports the measurements. Proofs of our main results, discussions and limitations, as well as additional supporting empirical results are available in the appendices.

2 Problem Formulation

This section describes the problem formulation: we state assumptions, form this paper's notations and introduce conceptual machineries that enable our results.

2.1 Notations and Preliminaries

Let $\mathcal{N} = \{1, \dots, N\}$ denote players' indices. For each player i, let x_i denote its action in a nonempty compact convex set $X_i \subseteq \mathbb{R}^{n_i}$. Player i observes other players' actions $x_{-i} = (x_j)_{j \neq i}$ and incurs a local cost $J_i : X \to \mathbb{R}$ with $J = (J_1, \dots, J_N)$ as a game's cost profile. The joint action set is the product $X = \prod_{i \in \mathcal{N}} X_i \subseteq \mathbb{R}^d$, $d = \sum_{i \in \mathcal{N}} n_i$, and $X_{-i} = \prod_{j \neq i} X_j$. The action profile $x = (x_1, \dots, x_N) \in X$ is also written as $x = (x_i, x_{-i})$ when player i's action is singled out. *We reserve $x^\star$ for an equilibrium and $\hat{x}$ for an arbitrary strategy profile at which a certificate is evaluated.* For vectors $v_1, \dots, v_N$ we write $\mathrm{col}(v_1, \dots, v_N)$ for the column vector obtained by stacking them. For a closed convex set S, the Euclidean projection $\Pi_S(x) := \arg\min_{y \in S} \|x - y\|$ is the unique element of S nearest x; since X is a product, $\Pi_X = (\Pi_{X_1}, \dots, \Pi_{X_N})$ acts blockwise. For differentiable J_i we write $\nabla_{x_i} J_i(x)$ for its gradient in player i's own action.

We now state the assumptions that underlie our theoretical machinery.

Assumption 2.1 (Compact convex product actions)**.** Each action set X_i is nonempty, compact, and convex, and the joint action set is the product $X = \prod_{i \in \mathcal{N}} X_i$.

Assumption 2.2 (Playerwise convexity)**.** For each player i and each $x_{-i} \in X_{-i}$, the cost $J_i(\cdot, x_{-i})$ is convex and continuously differentiable on X_i. The *pseudo-gradient* stacks the players' own-action gradients,

$$\Phi(x) \triangleq \mathrm{col}\left(\Phi_1(x), \dots, \Phi_N(x)\right) \in \mathbb{R}^d, \qquad \Phi_i(x) \triangleq \nabla_{x_i} J_i(x), \tag{1}$$

so that $\Phi_i(x) \in \mathbb{R}^{n_i}$ is player i's block of the pseudo-gradient.

We assume no monotonicity of the pseudo-gradient (1) and no map contraction. The product structure in Assum. 2.1 makes Π_X act blockwise and allows variation of a player's action under other players' fixed actions. Games whose feasibility sets couple across players fall outside the development.

All action coordinates and costs are nondimensionalized before applying the maps below: the unit of each action block is the chosen reference action scale, and each cost is divided by a fixed reference cost so that x_i and $\Phi_i(x)$ have the same numerical units. Thus $x - \Phi(x)$ uses a unit proximal step. With physical units, every occurrence of Φ can equivalently be replaced by $\gamma\Phi$ for a stated step size γ; the paper fixes this normalization to $\gamma = 1$.

Players' regret measures a profile's quality. The *exploitability* of $x \in X$, equivalently the max-over-players variant of the Nikaido–Isoda gap (Nikaidô & Isoda, 1955), whose original form sums over players rather than maximizing, is

$$\varepsilon(x) \triangleq \max_{i \in \mathcal{N}} \Big[J_i(x) - \min_{y_i \in X_i} J_i(y_i, x_{-i}) \Big], \tag{2}$$

in which each bracket is nonnegative, since $x_i \in X_i$ is feasible for the inner minimization. Hence $x \in X$ is a Nash equilibrium precisely when $\varepsilon(x) = 0$, and $\varepsilon(x)$ is the smallest tolerance for which x is an ε-Nash equilibrium (Radner, 1980).

Definition 2.1 (Variational inequality)**.** Let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty and let T map $\mathbb{R}^d$ into itself. The *variational inequality problem*, denoted $\mathrm{VI}(\mathcal{K}, T)$, is to find a vector $x^\star \in \mathcal{K}$ such that

$$T(x^\star)^\top (y - x^\star) \geq 0 \quad \text{for all } y \in \mathcal{K}. \tag{3}$$

Geometrically, (3) asks that $T(x^\star)$ make an acute angle with every feasible direction emanating from $x^\star$ (Harker & Pang, 1990). Under Assumption 2.1 and 2.2 the equilibria coincide with the solutions of $\mathrm{VI}(X, \Phi)$ (Rosen, 1965; Facchinei & Pang, 2010). The *natural map* of $\mathrm{VI}(\mathcal{K}, T)$ is $x \mapsto x - \Pi_{\mathcal{K}}\big(x - T(x)\big)$; it vanishes exactly at the solutions of (3), and we call its norm the *natural residual* (Facchinei & Pang, 2003, §1.5).

Definition 2.2 (Learned Response Map)**.** Given a dataset of players' observed actions, the response map of player i composes an *observer* h_i, which summarizes the other players' actions in a feature vector $h_i(x_{-i})$; a *response head* $g_i(\cdot\,; \theta_i)$, a function approximator with parameters θ_i; and a projection onto player i's action set,

$$f_i(x_{-i}; \theta_i) = \Pi_{X_i}\big(g_i(h_i(x_{-i}); \theta_i)\big). \tag{4}$$

Writing $F(x) = \mathrm{col}\,\big(f_1(x_{-1}), \ldots, f_N(x_{-N})\big)$ for the joint response of all players, a *behavioral equilibrium* is a profile $x^\star \in X$ for which every player's prediction reproduces its own input,

$$x_i^\star = f_i(x_{-i}^\star; \theta_i) \quad \text{for all } i \in \mathcal{N}, \tag{5}$$

equivalently $x^\star = F(x^\star)$ (Scarabaggio et al., 2024).

Because Π_{X_i} maps into X_i, we have $F(\mathbb{R}^d) \subseteq X$ and in particular $F : X \to X$, and because the projection onto a closed convex set is nonexpansive (Facchinei & Pang, 2003, §1.5), continuity of f_i reduces to continuity of the learned part $g_i \circ h_i$. Under Assum. 2.1 the joint action set is nonempty, compact and convex, so that Brouwer's theorem (Brouwer, 1911) supplies a profile satisfying (5) whenever each $g_i \circ h_i$ is continuous.

Suppose we write $h(x) = \mathrm{col}\,\big(h_1(x_{-1}), \ldots, h_N(x_{-N})\big)$ and $g = \mathrm{col}\,\big(g_1(\cdot\,; \theta_1), \ldots, g_N(\cdot\,; \theta_N)\big)$, so that $g(h(x))$ is the blockwise pre-projection output of (4). The *induced operator*

$$G(x) \triangleq x - g(h(x)), \tag{6}$$

satisfies $x - G(x) = g(h(x))$, and since Π_X acts blockwise under Assum. 2.1 this gives $F(x) = \Pi_X\big(x - G(x)\big)$. The fixed-point residual $x - F(x)$ of a projected learned map is therefore the natural-map residual of $\mathrm{VI}(X, G)$ (Facchinei & Pang, 2003, §1.5). The learned map is connected to the game via (i) the fixed-point residual of F, which one forward pass of the model computes; (ii) the operator-consistency error between (6) and (1), which cannot be formed without evaluating the true costs; and (iii) the natural residual of the variational inequality $\mathrm{VI}(X, \Phi)$. Only (i) can be computed without access to the game; any certificate that uses (ii) or (iii) must query the environment.

Remark 2.1 (A training loss does not bound δ). Small prediction loss leaves the operator-consistency error uncontrolled: a response map fitted to observed actions constrains G where data was observed, while Φ enters the exploitability (2) through its values on the region being certified. Bergerault et al. (2026) make the analogous point for multi-agent imitation learning: they exhibit games in which exact measure matching still leaves a large Nash gap, and prove that characterizing the Nash gap at a fixed measure-matching error is hard, recovering bounds only under additional structure on the expert equilibrium.

3 An Amortized Exploitability Certificate

Player i's regret is controlled by its blockwise residual and its gradient magnitude, both defined in (7). Lemma 3.1 establishes the pointwise residuals with local constants for the true pseudo-gradient. Lemma 3.2 produces a learned map, trading the variational residual for the fixed-point residual ρ_i (see Fig. 1) and the consistency error δ_i of (9). Every result in this section is proved in Appendix C, and Appendix D collects what the constants and the rate do and do not give: what an average prediction error fails to control, whether the coefficients can be shrunk, and when the rate improves from linear to quadratic.

3.1 A BLOCKWISE GAP BOUND

For a profile $x \in X$ write the gradient magnitude, blockwise residual, and the diameter of each player's block as

$$M_i(x) \triangleq \|\Phi_i(x)\|, \;\; D_i \triangleq \operatorname{diam} X_i, \;\; r_i(x) \triangleq \big\|x_i - \Pi_{X_i}\big(x_i - \Phi_i(x)\big)\big\|. \tag{7}$$

Here $\operatorname{diam} X_i = \sup_{u,v \in X_i} \|u - v\|$, finite by Assum. 2.1. Each $r_i(x)$ is the norm of block i of the right-hand side of the projected pseudo-gradient flow $\dot{x} = \Pi_X\big(x - \Phi(x)\big) - x$, so that every $r_i(x)$ vanishes exactly at the rest points of that flow: where the flow moves slowly, no player gains much by deviating.

Lemma 3.1 (Blockwise gap bound)**.** *Suppose that Assumption 2.1 and 2.2 hold. Then for every $x \in X$,*

$$\varepsilon(x) \;\le\; \max_{i \in \mathcal{N}} \; \min\big\{\big(M_i(x) + D_i\big)\, r_i(x), \;\; M_i(x)\, D_i\big\}. \tag{8}$$

Lemma 3.1 follows from the projection inequality (Facchinei & Pang, 2003, §1.5). Its constants $M_i(x)$ and D_i are local and blockwise: M_i is evaluated at x rather than maximized over X, and each refers only to player i's own gradient and own action set. Error bounds of this kind ordinarily control the distance from x to the solution set, for which the constant exists only under strong monotonicity (Yamashita & Fukushima, 1997). Equation (8) controls exploitability instead, for which compactness suffices. The second entry of the minimum, $M_i(x)D_i$, refers to no learned quantity and is available without a model, so that a certificate is informative only when it falls below $\max_i M_i(x)D_i$ — the comparison Corollary 3.1 makes precise.

3.2 TRANSFER TO A LEARNED MAP

Write $G_i(x) = x_i - g_i(h_i(x_{-i}))$ for the i-th block of the induced operator (6), and define the blockwise residual, ρ_i, and consistency error between the environment and learning scheme, δ_i as

$$\rho_i(x) \triangleq \big\|x_i - f_i(x_{-i})\big\|, \qquad \delta_i(x) \triangleq \big\|G_i(x) - \Phi_i(x)\big\|, \tag{9}$$

respectively.

Lemma 3.2 (Blockwise residual transfer)**.** *Let Assum. 2.1 hold and let Φ_i be defined on X. Then $r_i(x) \le \rho_i(x) + \delta_i(x)$ for every $x \in X$ and every player i.*

Proposition 3.1 (The learned map cannot improve a pointwise certificate where the projection is inactive)**.** *Let Assum. 2.1 hold and let the envelope of (12) be evaluated pointwise at $\hat{x}$, that is $\Omega = \{\hat{x}\}$, so that $\bar{M}_i = M_i(\hat{x})$ and $\bar{\delta}_i = \delta_i(\hat{x})$. Suppose that for every player i the projection in r_i is inactive at $\hat{x}$, that is $\hat{x}_i - \Phi_i(\hat{x}) \in X_i$, so that $r_i(\hat{x}) = M_i(\hat{x})$, and that $M_i(\hat{x}) > 0$. Then for every player i,*

$$\big(M_i(\hat{x}) + D_i\big)\big(\rho_i(\hat{x}) + \delta_i(\hat{x})\big) \;\ge\; M_i(\hat{x})^2 + M_i(\hat{x})D_i \;>\; M_i(\hat{x})D_i, \tag{10}$$

so at such a profile, each player's minimum in (8) is attained by the model-free entry and the certificate reduces to $\max_i M_i(\hat{x})D_i$.

Since forming $\delta_i(\hat{x})$ requires $\Phi_i(\hat{x})$, which already yields the model-free entry, the residual entry is worth its cost only when the consistency error is bounded once over a region rather than at the profile.

Scope of the pointwise limitation. Proposition 3.1 assumes every player's projection is inactive and $M_i(\hat{x}) > 0$; an active projection makes $r_i(\hat{x}) < M_i(\hat{x})$ and leaves the comparison open. The claim is that in the interior regime one environment query already supplies the model-free branch, which the learned branch cannot improve. Our measurements lie in that regime throughout, so they exhibit no pointwise improvement rather than showing none exists.

3.3 AMORTIZING THE ENVELOPE OVER A REGION

The consistency error is a property of the model against the environment, so it can be bounded once on a region and reused. The residual is a property of the model alone, so it is free wherever it is evaluated. Separating the two is what makes the certificate worth computing.

Definition 3.1 (Regional envelope)**.** Let $\Omega \subseteq X$ be closed. An *envelope* of a trained model on Ω is a pair of vectors $(\bar{M}, \bar{\delta})$ with, for every player i,

$$\bar{M}_i \geq \sup_{x \in \Omega} M_i(x), \qquad \bar{\delta}_i \geq \sup_{x \in \Omega} \delta_i(x). \tag{11}$$

Theorem 3.1 (Amortized exploitability certificate)**.** *Let Assumption 2.1 and 2.2 hold, let $(\bar{M}, \bar{\delta})$ envelope a trained model on $\Omega \subseteq X$, and let F be of the form (4). Then* every *profile $\hat{x} \in \Omega$ satisfies*

$$\varepsilon(\hat{x}) \;\leq\; \max_{i \in \mathcal{N}} \; \min\big\{ \big(\bar{M}_i + D_i\big)\big(\rho_i(\hat{x}) + \bar{\delta}_i\big),\;\; \bar{M}_i D_i \big\}, \tag{12}$$

and the residuals $\rho_i(\hat{x})$ require no access to the environment.

When $\bar{\delta}$ is measured by probes rather than supplied by an oracle, (12) inherits the confidence of that measurement and holds with probability at least $1 - \alpha$ (Proposition 4.1); the same qualification attaches to Corollary 3.1.

Corollary 3.1 (Non-vacuity threshold)**.** *Fix a player i and suppose $\bar{M}_i + D_i > 0$, which holds whenever X_i has more than one point. Then the residual entry of (12) improves on the model-free entry if and only if*

$$\rho_i(\hat{x}) + \bar{\delta}_i \;<\; \frac{\bar{M}_i D_i}{\bar{M}_i + D_i} \;\leq\; \min\{\bar{M}_i,\, D_i\}. \tag{13}$$

The threshold (13) is what player i's residual entry must improve on over the model-free one; Theorem 3.1 holds either way. Because (12) reports a maximum over players, the certificate improves on $\max_i \bar{M}_i D_i$ only when (13) holds at every player attaining that maximum; clearing it for some other player buys nothing. It is built from $\bar{M}_i$ and D_i, which carry no learned quantity, hence training cannot relax it, and a lower training loss need not lower $\bar{\delta}_i$ either (Proposition D.1). Since $\rho_i \geq 0$, the envelope alone must clear the threshold, at an accuracy fixed by the game rather than by the model; through (16) that accuracy in turn fixes the covering radius η, and the covering size grows as $(\operatorname{diam}\Omega/\eta)^{\dim\Omega}$.

Proposition 3.2 (A spurious stationary profile)**.** *Let $X_i = [-c, c]$ with $c > (1-\kappa)^{-1/2}$ and let*

$$J_i(x_i, x_{-i}) = \tfrac{1}{4}x_i^4 - \tfrac{1}{2}w_i(x_{-i})\,x_i^2, \qquad w_i(x_{-i}) = 1 + \tfrac{\kappa}{N-1}\sum_{j \neq i} x_j^2, \tag{14}$$

with $\kappa \in [0, 1)$. Then $\Phi(0) = 0$, hence $r_i(0) = 0$ for every player, while $\varepsilon(0) = \frac{1}{4}$.

The origin is a strict local maximum of each $J_i(\cdot, x_{-i})$ in its own argument, and it is the conditional mean of play whenever the data-generating process is symmetric across the two minimizers $\pm\sqrt{w_i(x_{-i})}$. A regressor fitted to conditional means therefore lands where the blockwise residual vanishes and exploitability is nonetheless $\frac{1}{4}$: without playerwise convexity, a vanishing residual certifies nothing. Under Assum. 2.2, multimodal play arises from multiple equilibria rather than from nonconvexity.

4 Envelope Costs

This section states the costs of the envelope in the sense of Definition 3.1 and the weaker guarantees available when those costs are omitted.

4.1 Probing at a profile

Two access models bracket what a probe can learn. A *gradient oracle* returns $\Phi_i(x)$ directly; a *cost oracle* returns only the values $J_i(x)$, so that Φ_i must be estimated. We assume the cost oracle, the access model of payoff-based equilibrium learning (Tatarenko & Kamgarpour, 2024). Differencing player i's own coordinates at radius $\tau > 0$ gives

$$\big[\widehat{\Phi}_\tau(x)\big]_{i,k} = \frac{J_i(x + \tau e_{i,k}) - J_i(x - \tau e_{i,k})}{2\tau}, \tag{15}$$

in which $e_{i,k}$ is the unit vector along coordinate k of player i's block. Only own coordinates are differenced, since $\Phi_i = \nabla_{x_i} J_i$ involves no cross-partial, and the estimator costs $2d$ evaluations per profile, with $d = \sum_{i \in \mathcal{N}} n_i$. Costs returned by a simulator, or averaged over finitely many rollouts of a stochastic environment, are not exact, so we take a query at x to return $J_i(x) + Z$ with Z independent across queries, mean-zero and σ-sub-Gaussian. Equation (15) then carries both a smoothing bias and a noise term, and τ trades one against the other.

Remark 4.1 (Probes at the boundary). Since every f_i ends in a projection onto X_i, certified profiles frequently lie on ∂X, and then $x \pm \tau e_{i,k}$ leaves X where the costs are neither defined nor assumed smooth. Two remedies are available. Proposition 4.1 adopts the first, assuming each J_i extends three times continuously differentiably to a τ-neighborhood of X, letting its Taylor expansion run along the whole probe segment. When unavailable, we replace (15) by a one-sided difference on coordinates within τ of the boundary. This has bias $O(B_2\tau)$, with B_2 bounding the second derivative. Thus, the balanced radius on the coordinates becomes $\tau \asymp (\sigma/B_2)^{1/2}$ and their contribution to the penalty is $O\big(\sigma^{1/2} B_2^{1/2}\big)$, in both cases up to the $\sqrt{n_i}$ and logarithmic factors that (16) carries.

4.2 From probes to a regional envelope

A bound on $\sup_{x \in \Omega} \delta_i(x)$ from finitely many probes needs a modulus of continuity. Let δ_i be L_δ-Lipschitz on Ω, which holds whenever both Φ_i and the learned head are Lipschitz there.

Proposition 4.1 (Envelope from a finite covering)**.** *Let $\mathcal{Z} \subseteq \Omega$ be a finite η-covering of the certified region Ω, let δ_i be L_δ-Lipschitz on Ω, let each $J_i(\cdot, x_{-i})$ extend three times continuously differentiably to a τ-neighborhood of X with $|\partial^3 J_i / \partial x_{i,k}^3| \le B_3$ there, and let each cost evaluation be corrupted by independent, mean-zero, σ-sub-Gaussian noise. Then for any $\alpha \in (0,1)$, with probability at least $1 - \alpha$, the quantities*

$$\bar{\delta}_i \;=\; \max_{z \in \mathcal{Z}} \big\| G_i(z) - \widehat{\Phi}_{\tau,i}(z) \big\| \;+\; L_\delta \eta \;+\; \tfrac{1}{6} B_3 \tau^2 \sqrt{n_i} \;+\; \tfrac{\sigma}{\tau} \sqrt{n_i \log(2d|\mathcal{Z}|/\alpha)} \qquad (16)$$

satisfy the second half of (11). If in addition M_i is L_M-Lipschitz on Ω, the same construction with L_M in place of L_δ bounds $\bar{M}_i$.

Balancing the bias and noise terms in (16) at $\tau \asymp \big(\sigma \sqrt{\log(2d|\mathcal{Z}|/\alpha)}/B_3\big)^{1/3}$ leaves a penalty of order $\sqrt{n_i}\,\big(\log(2d|\mathcal{Z}|/\alpha)\big)^{1/3} \sigma^{2/3} B_3^{1/3}$, which vanishes with the oracle noise. The covering term $L_\delta \eta$ is a modeling cost, and $|\mathcal{Z}|$ grows by $(\operatorname{diam}\Omega/\eta)^{\dim \Omega}$, so that a rigorous envelope is affordable only when Ω is low-dimensional. The regions for which this is practical are therefore structured ones: a line segment between two candidate profiles, the trajectory of a response iteration, or a neighborhood of a single deployment point.

Remark 4.2 (What the envelope assumes about the environment). Proposition 4.1 needs a Lipschitz modulus L_δ for the consistency error, a third-derivative bound B_3 on costs that are by assumption unknown, a noise scale σ, and L_M for the gradient envelope. None is playerwise and none is measurable from cost-oracle queries alone, so that a global hypothesis on the *model*, that it contracts, has been exchanged for global hypotheses on the *environment*. The exchange is worth making because on a structured low-dimensional Ω these constants are falsifiable: a probe that violates a posited L_δ or B_3 exhibits the violation, whereas a contraction modulus asserted of a trained network is not checkable at all.

5 Experiments

We test soundness, the pointwise limitation, and regional amortization. Exploitability is computed from the analytic game, not the learned response. The seven responders are an affine baseline, two-layer 32–32 tanh/ReLU, spectral-normalized and Almost-Orthogonal-Layer (AOL) Lipschitz-certified networks (Prach & Lampert, 2022), and 64–64 conditional diffusion and flow models. Standard E1–E2 runs supply 240 noisy observational transitions; neural responders use Adam with batch size 128 for 200 epochs at learning rate 3×10^{-3} (AOL: 5×10^{-3}), with least-squares, denoising, and flow-matching losses as appropriate. E8 uses its stated reduced training budget. A cell is one game, response family, and seed; comparisons are paired within cells, with 10,000-resample bootstrap intervals (Efron, 1979), Wilcoxon signed-rank tests (Wilcoxon, 1945) and Holm's step-down correction (Holm, 1979). The full protocol appears in Appendix F.

Bound and Tightness of Learned Map. The tightness study fits seven response families over 20 seeds on three games: a coupled quadratic game whose coupling parameter fixes the monotonicity modulus exactly, a capacitated Cournot game whose capacities bind at equilibrium, and a Rosen game carrying one constraint shared across players. The quadratic and Cournot games have product action sets and test Theorem 3.1 directly: $2 \times 7 \times 20 = 280$ cells, all valid. The Rosen game violates the product requirement of Assum. 2.1; its 140 cells are reported only as a stress diagnostic and are excluded from theorem coverage. The shared game has an analytically specified best-response oracle, but its variational equilibrium is computed to the documented numerical tolerance rather than in closed form.

The learned-map-free arm is active in every cell, with median tightness 108.7; the residual arm has median 297.9(about 2.7 times larger). Its paired difference from the direct-probe comparison is -0.001165in $\log_{10}$ tightness ($p_{\mathrm{Holm}} = 0.1519$); the test does not detect a difference. The learned response helps only after the consistency error is audited over a region and reused.

Playerwise scaling. The scaling study varies the player count while holding the observational-data budget fixed. At a fixed displacement from the reference equilibrium, the aggregate ratio of certificate to true exploitability increases from 38.15 at $n = 2$ to 1157 at $n = 256$. The corresponding blockwise ratio changes from 13.92 to 6.628. The aggregate ratio therefore grows by a factor of about thirty, whereas the blockwise ratio remains on the same order, which Fig. 2 shows at both action dimensions and Table 1 lists at scalar actions. Shading marks where the aggregated bound exceeds the model-free bound; the dash-dot rule marks where the responder fit becomes underdetermined.

The aggregate form uses one global constant and a joint residual, whereas the playerwise form uses (7); here the blockwise series equals the model-free entry. The fixed 240-transition budget makes the largest fits underdetermined, which can inflate learned-map residuals but not the model-free arm.

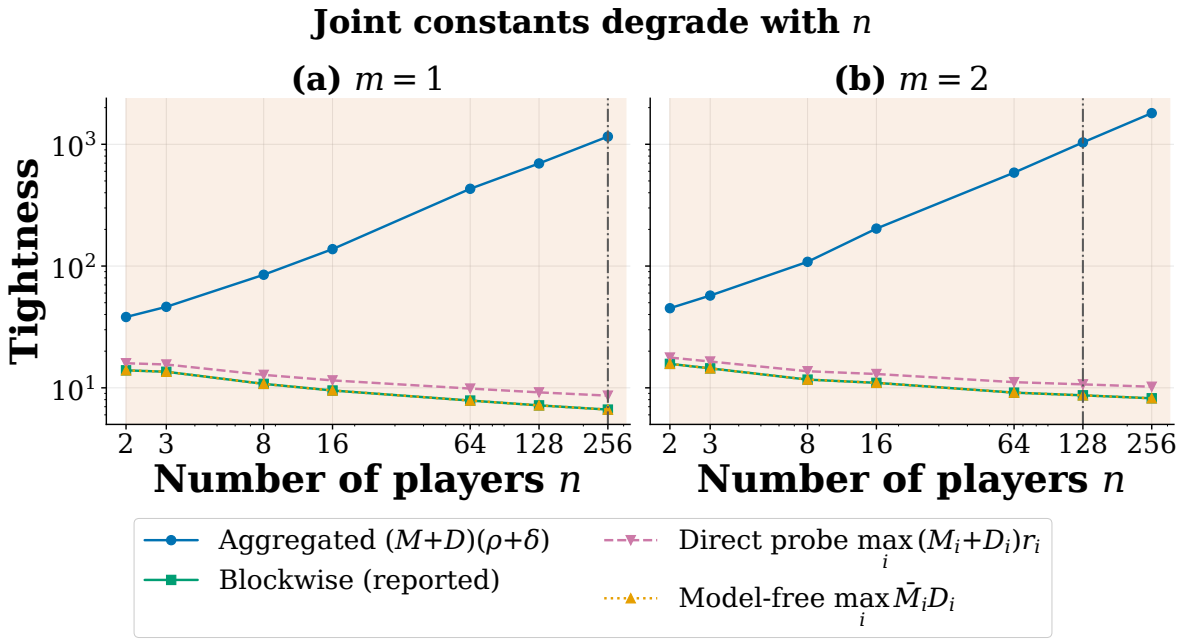


Fig. 2: **Joint constants degrade with the player count; playerwise ones do not.** Coupled quadratic game, (a) scalar and (b) two-dimensional actions, medians over 20 seeds, log–log, lower is tighter.

Regional auditing creates the amortization regime. At $\eta / \operatorname{diam} \Omega = 0.025$, the one-dimensional segment requires 20 covering points and the three-dimensional box requires 8000. Their residual arms bind on 0.5742 and 0.7266 of held-out profiles, with regional-to-pointwise ratios 1.788 and 1.725. Every tested profile satisfies the audited envelope; covering cost grows with region dimension. Fig. 3 shows the tradeoff.

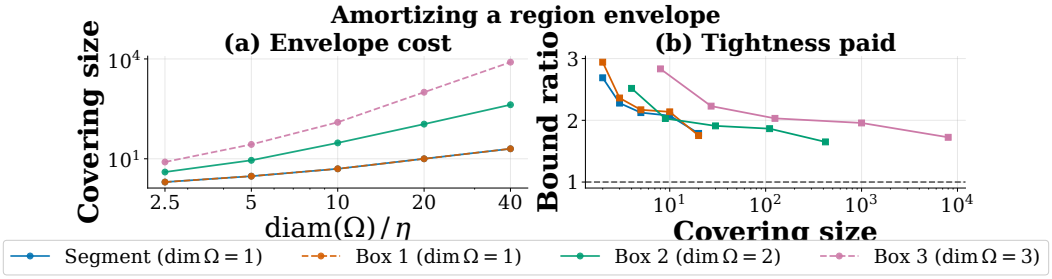


Fig. 3: **Regional envelope costs and tightness.** Medians over 10 seeds and 256 held-out profiles per region. (a) Covering size=break-even certified profile numbers. (b) Amortized bound over pointwise bound; the dashed rule is the value a single-profile probe attains.

Contraction and exploitability measure different properties. Pooling the theorem-supported product-set cells from the tightness study with the coupling sweep gives 1000 model cells; the shared-constraint diagnostics are excluded. Small-gain certification accepts 0.09, while 0.524 have exploitability at most 0.01. The four contingency counts are 70, 20, 454, and 456; contraction is therefore neither necessary nor sufficient here. Generative families are excluded because they have no rigorous global Lipschitz bound.

Multiplicity, solvers, and probe diagnostics. On the playerwise-convex non-monotone game, generative responders are evaluated through conditional-mean maps, so the comparison is silent about generativity and capacity; their heads have roughly four times as many parameters yet recover fewer equilibria (Appendix E). The nonconvex witness remains a scope diagnostic: zero variational residual can coexist with nonzero exploitability. On the nonexpansive game, Picard reaches tolerance 10^{-8} in none of 20 runs, while Krasnosel'skiĭ–Mann and Halpern require medians of 126 and 5 iterations. Probe-radius sweeps reproduce the bias–noise tradeoff of Proposition 4.1; both are diagnostics, not replacements for its regularity assumptions.

6 Conclusion

A contraction certificate establishes the regularity of a learned response map without resolving its fidelity to the game that produced the data. We address this with a blockwise bound on exploitability in the true game, whose hypotheses constrain the environment rather than the model. Two limits bound what it delivers: Proposition 3.1 shows the learned map cannot improve a certificate at a single probed profile where the projection is inactive, and Corollary 3.1 shows the regional envelope that escapes that argument must reach an accuracy set by the game, at a cost exponential in $\dim \Omega$. Proposition 3.2 isolates the indispensable hypothesis: without playerwise convexity a vanishing residual coexists with exploitability $\frac{1}{4}$. Empirically, contraction certification and low exploitability come apart. Appendix B discusses our limitations.

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APPENDICES

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A Related Works and Distinctions

The Case for Certification No uncoupled dynamic converges to Nash equilibrium across a family of games containing a neighborhood of a given game (Hart & Mas-Colell, 2003), and their witness lies inside the hypotheses adopted here: two players on unit disks, costs smooth and strongly convex in each player's own action, a unique strict equilibrium. As stated in Hart & Mas-Colell (2003), converging to equilibrium is a more stringent requirement than being in equilibrium. Certificates address being in equilibrium.

Learned response maps. Response-based models of interaction replace a player's optimization with a map fitted to its observed actions. Scarabaggio et al. (2024) introduced the projected observer–head architecture, established via Brouwer's theorem that a fixed point exists, and showed that such a fixed point solves a variational inequality; uniqueness and convergence of the response iteration follow from an aggregate contraction condition $\left(\sum_i \gamma_i^2\right)^{1/2} < 1$, relaxed to a Krasnosel'skiĭ iteration at the nonexpansive boundary. Their guarantees concern the learned game. The distinction drawn in this paper is that Theorem 3.1 bounds exploitability in the game that generated the data, at a profile that need not be an exact fixed point, and needs no condition on γ_i. A subsequent treatment from the same group combines projected maps, existence, and aggregate contraction in the same way (Mignoni, 2025, Chapter 6). Their variational inequality is stated in the learned map and is equivalent to the fixed-point condition itself, so that it constrains the game that generated the data nowhere. Discarding the learned component, (Scarabaggio et al., 2026) characterizes stationary profiles of the true costs as fixed points of the blockwise projected gradient and establish existence and homotopy persistence by Lefschetz–Nielsen theory. We measure the blockwise residual r_i of the map.

A different ε. In the purification of Harsanyi (1973) the parameter scales a perturbation of the payoffs and the equilibrium is exact in the perturbed game; in quantal response (McKelvey & Palfrey, 1995) and in perturbed best-response learning (Hofbauer & Hopkins, 2005) it scales a smoothing and the equilibrium is an exact fixed point of a *modified* operator, whose exploitability in the original game is in general bounded away from zero. The certificate here bounds regret in the unmodified game, so convergence to a perturbed equilibrium implies nothing about (2), and conversely.

Decision-dependent and performative games. When the data distribution reacts to the deployed model, the fixed point of repeated retraining becomes the solution concept. Perdomo et al. (2020) introduced performative prediction and the repeated-risk-minimization fixed point, and Narang et al. (2023) extended it to several players, proving existence, uniqueness, and a linear rate for performatively stable equilibria when a sensitivity modulus divided by a strong-convexity modulus falls below one. That ratio plays the role q plays here, and the resulting sensitivity bound carries the same $1/(1-\cdot)$ constant that Remark B.3 dispenses with. Outside the contractive regime the dynamics are not merely unanalyzed but genuinely irregular: Piliouras & Yu (2023) exhibit chaotic multi-agent retraining, and Dong et al. (2023) study regions of attraction where contraction holds only locally. These results motivate a certificate that does not degrade as the modulus approaches one.

Certified operators and Lipschitz networks. A parallel literature constrains what a network can do rather than enveloping what it did. Monotone operator equilibrium networks obtain well-posed implicit layers by construction (Winston & Kolter, 2020), and Jafarpour et al. (2021) certify implicit networks through a non-Euclidean, weighted one-sided Lipschitz condition that is strictly weaker than a Euclidean contraction; Davydov et al. (2022) develop the contraction theory behind it. Lipschitz bounds are enforced architecturally by direct parameterization (Wang & Manchester, 2023) and unified across several such constructions by Havens et al. (2023), or estimated post hoc by semidefinite relaxation (Fazlyab et al., 2019), exact computation being NP-hard (Virmaux & Scaman, 2018). Two cautions from this literature shape our treatment. Products of layerwise spectral norms bound the joint operator norm only loosely, and the Clarke chain rule for a composition holds as an inclusion rather than an equality (Clarke, 1983), so a layerwise regularity certificate is unsound. Automatic differentiation compounds the problem: it returns an element of a conservative field rather than of the Clarke Jacobian, and the two differ on activation and projection boundaries (Bolte & Pauwels, 2021). We therefore report sampled Jacobian quantities as lower-bound diagnostics and never as certificates.

Error bounds and gap functions. The inequality behind Lemma 3.1 is part of the error-bound theory for finite-dimensional variational inequalities (Facchinei & Pang, 2003, Ch. 6)(Pang, 1997). Gap and merit functions convert a variational inequality into an optimization problem and supply computable residuals: the regularized gap of Fukushima (1992), the gap-function class of Larsson & Patriksson

(1994), the D-gap of Peng (1997), the global error bounds of Yamashita & Fukushima (1997), and the Nikaido–Isoda reformulations for generalized games of von Heusinger & Kanzow (2009). These results run predominantly from a residual to *distance-from-solution* and require strong monotonicity for the constant to exist; our use runs instead to *exploitability*, which is why compactness suffices. Sensitivity of solutions under perturbation of the defining map is classical, through Robinson's strong regularity (Robinson, 1980), its characterization on polyhedral sets (Dontchev & Rockafellar, 1996), and the bound of Dafermos (1988) in which the contraction modulus is replaced by a strong-monotonicity modulus of the environment.

Certifying a limit vs. a profile. Mertikopoulos & Zhou (2018) work in our setting, with compact convex product action sets and payoffs only C^1 in each player's own action. However, what they certify is a limit point of dual averaging which is Nash only if the iterates converge and the game is pseudo-concave. Their global results require that the Nash set be variationally stable, the pseudo-gradient be monotone, or the game to admit a concave potential. Their one quantitative gap estimate is an $\mathcal{O}(1/\sqrt{n})$ rate for an ergodic average that they note may behave differently from the played sequence. Variational stability is adoptable and would furnish isolation and asymptotics. However, it cannot furnish a certificaten since it relates $\Phi(x)$ to an unknown equilibrium. Thus, with that equilibrium unknown it yields no computable number, and verifying it requires the game to be Hessian at the point one is trying to locate. For C^2 costs it is moreover equivalent to local asymptotic stability (Mazumdar et al., 2020), so that it excludes precisely the equilibria that gradient-based learners avoid from almost every initialization — the profiles most in need of an external certificate.

Continuous-time equilibrium seeking. A parallel literature drives dynamics to an equilibrium rather than certifying a profile. The survey of Hu et al. (2022) collects the consensus-based, passivity-based and gradient-free designs, and Belgioioso et al. (2022) give the operator-splitting counterpart for coupled constraints. Two features bear on this work. Their convergence certificates are Lyapunov functions of the form $\frac{1}{2}\|x - x^\star\|^2$, evaluable only once $x^\star$ is known, which is unavailable when the game is not; and their hypotheses begin at monotonicity of Φ and strengthen from there. Gap and merit functions for variational inequalities are cited in neither survey, and no ε-equilibrium notion appears in either. The single quantitative rate in Belgioioso et al. (2022) is an ergodic $\mathcal{O}(1/k)$ bound on a fixed-point residual, and their operator-to-game link holds at an exact zero of that residual, so that a positive residual is converted into a statement about the game nowhere in this line.

Identifying a game from observed play. Whether observed actions pin down a game is the question our consistency error confronts. Gürkan & Pang (2009) give conditions under which equilibria of approximating games converge to equilibria of the true game, via a multi-epiconvergence notion; the statements are asymptotic and contain no explicit constant, and the authors note the connection between their regularized Nikaido–Isoda function and variational gap functions without developing it quantitatively. Bemporad & Tatarenko (2026a) learn solution maps of parametric generalized games by training against the Nikaido–Isoda gap rather than regressing on precomputed equilibria, which requires only best-response data, and Bemporad & Tatarenko (2026b) instead secure monotonicity of the learned game by construction. Bergerault et al. (2026) prove the sharpest negative result in this direction: characterizing the Nash gap at a fixed measure-matching error is hard, and exact matching of observed play can still leave a large Nash gap. That result is why Remark 2.1 measures operator consistency at a point rather than distributional fit, and why the environment access of Section 4 is stated as a cost rather than assumed away. Payoff-based equilibrium learning supplies the access model we adopt, in which players evaluate costs but do not differentiate them (Tatarenko & Kamgarpour, 2024).

Generative responders. Our set-valued responders draw on conditional generative models, which represent multimodal conditional distributions rather than conditional means. Proposition 3.2 shows why that distinction is not cosmetic for equilibrium computation where playerwise convexity fails. Where it holds, our measurements in Section 5 attribute recovery of multiple equilibria to model capacity rather than to a multimodal likelihood; we therefore do not argue that generativity is required. We are not aware of prior work analyzing diffusion policies, flow matching, or vision–language–action models as response correspondences in a game, though we have not surveyed those literatures systematically and state this as an impression rather than a priority claim. Two scope limits apply when the certificate is read alongside them. Theorem 3.1 constrains the *environment* through Assum. 2.2, so that continuity of the learned map is not by itself sufficient; and under playerwise convexity each best-response set is convex, so the multimodality that motivates a generative responder must come from multiple equilibria rather than from a multimodal best response. Certifying a

genuinely set-valued response, where the residual is defined for a sampler rather than for a map, remains open, as does certifying a generalist policy deployed simultaneously as several players, whose playerwise responses share parameters.

B Discussion and Limitations

The envelope is the binding cost. A rigorous regional envelope requires a finite covering, and $|\mathcal{Z}|$ grows exponentially in $\dim \Omega$ (Section 4). This confines rigorous envelopes to structured low-dimensional regions: a segment between two candidate profiles, the trajectory of a response iteration, or a neighborhood of one deployment point. Sampling Ω in place of covering it scales, but certifies a randomly drawn profile rather than every profile in the region, so that the uniform conclusion is the thing given up. Closing that gap with a scalable uniform bound, whether through structure in the learned head or through a concentration argument over Ω, is the most useful extension we can identify.

Coupled feasibility sets are excluded. Assum. 2.1 requires a product action set, and the requirement is load-bearing twice: it makes the projection act blockwise, and it permits one player's action to be varied while the others are held fixed. Generalized games with shared constraints therefore fall outside every result here. Our coupled-constraint experiment shows what the certificate does when its own hypotheses are violated, and is not a claim of coverage.

Playerwise convexity is necessary, and it makes the set-valued question unanswerable here. Proposition 3.2 shows that the blockwise residual certifies nothing once Assum. 2.2 fails. The assumption also removes the motivation for generative responders: under playerwise convexity each best-response set is convex, so that responses of the bimodal kind in Proposition 3.2 are excluded by hypothesis. The multimodality that survives arises from multiple *equilibria* of a playerwise-convex game, which non-monotone coupling produces readily. Our evaluation on those games (Section 5) reports what a *conditional-mean* responder recovers, since the residual ρ_i of (9) requires a single-valued evaluation and the generative families expose one only through their mean map. It therefore speaks to neither generativity nor capacity: the comparison is between mean maps, and the generative heads carry roughly four times the parameters and still do worse. What it does show is that a linear regressor fitted to play pooled across basins lands between them on a fraction 0.724 of instances, while nonlinear responders do not. Whether a certificate can be given for a genuinely set-valued response we leave open, and the open half is the metric one rather than the topological one. Existence for a response correspondence with nonempty closed convex values is classical (Kakutani, 1941; Facchinei & Kanzow, 2010); what is missing is a *residual*, since no notion of approximate fixed-pointness of a correspondence appears in the equilibrium-seeking literature we surveyed, and ρ_i as defined here requires a single-valued evaluation.

The quadratic rate is not a computable certificate. Proposition D.2 governs the interior regime but requires distances to an $\arg\min$, hence n constrained convex solves in the unknown costs. We report it as a rate and measure it only where ground truth is available. A computable surrogate would need interiority certified from probe data together with a curvature modulus, which we have not established.

Sampled Jacobians are diagnostics, not certificates. Where the experiments report operator norms of learned maps, rigorous bounds are computed from weights and sampled quantities are labeled as lower-bound diagnostics. The distinction is not pedantic: the Clarke chain rule for a composition holds only as an inclusion (Clarke, 1983), and automatic differentiation returns an element of a conservative field rather than of the Clarke Jacobian, the two differing on activation and projection boundaries (Bolte & Pauwels, 2021). No certificate in this paper reads a sampled Jacobian.

Remark B.1 (Why certify rather than solve). The same oracle that supplies an envelope would also drive projected gradient play, so that the certificate must earn its place against simply solving the game. It does so on three counts. (i) An envelope is paid once and then certifies every profile in Ω for the price of a forward pass, whereas solving pays along every trajectory and yields one profile. (ii) Projected gradient play carries no convergence guarantee at our hypotheses, and the failure arrives one rung earlier than the absence of monotonicity: under monotone Φ the Lyapunov derivative is still nonpositive, but the set on which it vanishes gives only $\langle \Phi(x) - \Phi(x^\star), x - x^\star \rangle = 0$, which does not force $x = x^\star$, so the invariance argument does not close (Hu et al., 2022). Below monotonicity the dynamics are irregular rather than merely unanalyzed, and multi-player retraining outside the contraction regime admits chaotic orbits (Piliouras & Yu, 2023). (iii) The learned map is often the deliverable for reasons unrelated to equilibrium computation, such as simulating counterfactual play, so the question is whether the model one already holds is trustworthy where it is evaluated.

Remark B.2 (What the certificate does not claim). The bound (12) concerns profiles in Ω. It does not assert that $\mathrm{Fix}(F)$ is a singleton, nor that a certified profile approximates any particular equilibrium. For a nonexpansive F the fixed-point set is nonempty, closed, and convex, but need not be a singleton, as $F = \mathrm{Id}$ shows. We therefore write *a* behavioral equilibrium throughout, never *the* behavioral equilibrium.

Remark B.3 (Behavior at the contraction boundary). Classical fixed-point perturbation controls equilibrium displacement by $\varsigma/(1-q)$, for a uniform map error ς and a map of contraction modulus $q < 1$ (Bauschke & Combettes, 2017), and that constant diverges as $q \to 1$. The certificate (12) contains no q. Certifying a nonexpansive or expansive learned map is thus a regime the bound already covers, which is the operative difference from the aggregate-contraction condition of Scarabaggio et al. (2024) and from the sensitivity-to-curvature ratio governing performatively stable equilibria in Narang et al. (2023).

Remark B.4 (Regularity of the learned map certifies nothing). Let $F \equiv \hat{x}$ be constant for an arbitrary $\hat{x} \in X$. That map is nonexpansive and $\rho_i(\hat{x}) = 0$ for every player, while $\varepsilon(\hat{x})$ is as large as the game permits. Its induced operator satisfies $G_i(\hat{x}) = 0$, so $\delta_i(\hat{x}) = M_i(\hat{x})$ and the residual entry of (12) reads $(\bar{M}_i + D_i)\bar{M}_i$, which exceeds the model-free entry $\bar{M}_i D_i$. The constant map is therefore excluded by the model-free bound rather than by the envelope, and the certificate reports nothing better than the trivial bound for it. Regularity of a learned map carries no information about the game.

Remark B.5 (Remark on Proposition 4.1.). A strict contraction F with modulus $q < 1$ gives geometric convergence of the Picard iteration $x^{k+1} = F(x^k) = \Pi_X\big(x^k - g(h(x^k))\big)$. To reduce the fixed-point residual below ϵ, the number of value queries scales as

$$\mathcal{O}\left(\frac{\log(1/\epsilon)}{1-q}\right)$$

when q is close to one. Equivalently, if $q = 1 - \varepsilon$ with $0 < \varepsilon \ll 1$, the cost is $\mathcal{O}(\varepsilon^{-1}\log(1/\epsilon))$. At the nonexpansive boundary $q = 1$, this contraction argument disappears. General value-oracle lower bounds for Brouwer fixed-point computation are exponential in the dimension and in the number of digits of accuracy, under the corresponding Lipschitz assumptions (Hirsch et al., 1989); related reductions yield query lower bounds for equilibrium computation (Babichenko, 2016). These results provide complexity context, rather than a bound on the exploitability certificate (12): our result certifies a profile without assuming contraction, whereas Picard iteration seeks a fixed point under contraction.

Remark B.6 (A cheaper envelope with a weaker guarantee). Replacing the covering by N independent draws from a distribution ν on Ω removes both the exponential dependence on $\dim \Omega$ and the Lipschitz modulus L_δ, at the price of the conclusion. A sampled envelope bounds δ_i with high probability at a ν-random profile rather than uniformly on Ω, so that the resulting statement certifies a randomly drawn deployment and not every profile in the region. Applied as audited, the covering envelope of Proposition 4.1 dominated the pointwise constants at every one of 2560 held-out profiles on the three-dimensional box, and at every held-out profile in all twenty region–resolution cells. A sampled envelope over a region of the same game, built from a finite draw in place of a covering, failed to dominate at a fraction 0.0314 of fresh profiles, at which the region bound is therefore not a bound at all. We report the sampled construction as what it is, and never as a bound on the supremum.

The boundary-stencil count is not a measurement. The fraction of coordinates requiring the one-sided stencil of Remark 4.1 is 0 *by construction* rather than by measurement: the probe used in our runs is an unconstrained central difference that never switches stencil, so the number could not have come out otherwise. The companion quantity we can measure is the fraction of probe steps that leave the action set, which is 0 at every radius on the grid. A feasibility-respecting probe would therefore not have switched either, but that is a property of these arenas and not evidence about the remedy.

C A CATALOG OF (DEFERRED) PROOFS

This appendix catalogs the proofs of the results of Section 3 and Section 4, in order.

Proof of Lemma 3.1. Fix a player i and let $y_i^\star \in \arg\min_{y_i \in X_i} J_i(y_i, x_{-i})$, which exists because X_i is compact by Assum. 2.1 and $J_i(\cdot, x_{-i})$ is continuous by Assum. 2.2. Convexity of $J_i(\cdot, x_{-i})$ implies that $J_i(y_i^\star, x_{-i}) \geq J_i(x) + \langle \Phi_i(x), y_i^\star - x_i \rangle$. Hence, player i's regret obeys

$$J_i(x) - J_i(y_i^\star, x_{-i}) \;\leq\; \big\langle \Phi_i(x),\, x_i - y_i^\star \big\rangle. \tag{17}$$

Set $p_i = \Pi_{X_i}\big(x_i - \Phi_i(x)\big)$ and $w_i = x_i - p_i$, so that $\|w_i\| = r_i(x)$ by (7). Projecting $x_i - \Phi_i(x)$ onto X_i implies $\langle x_i - \Phi_i(x) - p_i,\, y_i - p_i \rangle \leq 0$ for every $y_i \in X_i$, the variational characterization of the projection (Facchinei & Pang, 2003, §1.5); equivalently, p_i solves $\mathrm{VI}\big(X_i,\, u \mapsto u - (x_i - \Phi_i(x))\big)$ in the sense of Definition 2.1. It is easy to verify that $\langle \Phi_i(x),\, p_i - y_i \rangle \leq \langle w_i,\, p_i - y_i \rangle$; applying the Cauchy–Schwarz inequality with $\|p_i - y_i\| \leq D_i := \operatorname{diam} X_i$, since $p_i, y_i \in X_i$, gives

$$\langle \Phi_i(x),\, p_i - y_i \rangle \;\leq\; r_i(x)\, D_i \qquad \text{for every } y_i \in X_i. \tag{18}$$

Splitting (17) at p_i as $\langle \Phi_i(x), x_i - y_i^\star \rangle = \langle \Phi_i(x), w_i \rangle + \langle \Phi_i(x), p_i - y_i^\star \rangle$, then bounding the first term by $M_i(x) r_i(x)$ with the Cauchy–Schwarz inequality and the second by (18), we find that the bound on the player i's regret is $\big(M_i(x) + D_i\big) r_i(x)$, which is the first entry of (8). Separately, (17) and Cauchy–Schwarz give $J_i(x) - J_i(y_i^\star, x_{-i}) \leq M_i(x) \|x_i - y_i^\star\| \leq M_i(x) D_i$. Both bounds apply to the same player, hence player i's regret is at most their minimum, and (8) follows by taking the maximum over players. Writing $a_i = \big(M_i(x) + D_i\big) r_i(x)$ and $b_i = M_i(x) D_i$, the inequality $\max_i \min\{a_i, b_i\} \leq \min\{\max_i a_i, \max_i b_i\}$ shows that taking the minimum after the maximum is valid but never tighter, and strict exactly when no single player has both $a_i \geq m$ and $b_i \geq m$ for $m = \min\{\max_j a_j, \max_j b_j\}$. □

Proof of Lemma 3.2. Because g and h stack blockwise, $x - G(x) = g(h(x))$ gives $[x - G(x)]_i = g_i(h_i(x_{-i}))$, so that $f_i(x_{-i}) = \Pi_{X_i}\big([x - G(x)]_i\big)$ by (4). Inserting $f_i(x_{-i})$ into $r_i(x) = \|x_i - \Pi_{X_i}(x_i - \Phi_i(x))\|$, the triangle inequality gives $r_i(x) \leq \|x_i - f_i(x_{-i})\| + \|f_i(x_{-i}) - \Pi_{X_i}(x_i - \Phi_i(x))\|$, whose first term is $\rho_i(x)$. Nonexpansiveness of Π_{X_i} bounds the second term by $\big\|[x - G(x)]_i - (x_i - \Phi_i(x))\big\| = \|\Phi_i(x) - G_i(x)\| \triangleq \delta_i(x)$. Hence, $r_i(x) \leq \rho_i(x) + \delta_i(x)$. □

Proof of Proposition 3.1. Because $\hat{x}_i - \Phi_i(\hat{x}) \in X_i$, the projection defining r_i fixes its argument, so $\Pi_{X_i}\big(\hat{x}_i - \Phi_i(\hat{x})\big) = \hat{x}_i - \Phi_i(\hat{x})$ and therefore $r_i(\hat{x}) = \|\Phi_i(\hat{x})\| = M_i(\hat{x})$. Lemma 3.2 gives $r_i(\hat{x}) \leq \rho_i(\hat{x}) + \delta_i(\hat{x})$, hence $\rho_i(\hat{x}) + \delta_i(\hat{x}) \geq M_i(\hat{x})$. Multiplying by $M_i(\hat{x}) + D_i \geq 0$ yields the first inequality of (10), and the second is strict whenever $M_i(\hat{x}) > 0$, since then $M_i(\hat{x})^2 > 0$.

Without the inactive-projection hypothesis the conclusion can fail. On $X_i = [0, 1]$ take $\hat{x}_i = 0$ and $\Phi_i(\hat{x}) = 1$, so that $\hat{x}_i$ is player i's constrained best response and its regret is zero. Then $\hat{x}_i - \Phi_i(\hat{x}) = -1 \notin X_i$, and a model with $G_i(\hat{x}) = \Phi_i(\hat{x})$ has $f_i(\hat{x}_{-i}) = \Pi_{X_i}(-1) = 0$, so $\rho_i(\hat{x}) = \delta_i(\hat{x}) = 0$ while $M_i(\hat{x}) = 1$. The residual entry is then 0 and the model-free entry is D_i, so the learned map does sharpen the certificate, and it does so where the projection is active. □

Proof of Theorem 3.1. Fix $\hat{x} \in \Omega$. By (11), $M_i(\hat{x}) \leq \bar{M}_i$ and $\delta_i(\hat{x}) \leq \bar{\delta}_i$, and Lemma 3.2 gives $r_i(\hat{x}) \leq \rho_i(\hat{x}) + \bar{\delta}_i$. Both entries of (8) are nondecreasing in M_i and in r_i on the nonnegative orthant $\mathbb{R}_+$ since $D_i \geq 0$. Hence, substituting the enveloped upper bounds preserves each inequality. Applying Lemma 3.1 playerwise gives (12). The envelope is fixed before $\hat{x}$ is chosen, so the conclusion holds simultaneously for every $\hat{x} \in \Omega$. □

Proof of Corollary 3.1. Dividing $(\bar{M}_i + D_i)(\rho_i + \bar{\delta}_i) < \bar{M}_i D_i$ by it gives the first inequality of (13). For the second, $\bar{M}_i D_i / (\bar{M}_i + D_i) \leq \bar{M}_i$ because $D_i \leq \bar{M}_i + D_i$, and $\bar{M}_i D_i / (\bar{M}_i + D_i) \leq D_i$ because $\bar{M}_i \leq \bar{M}_i + D_i$; hence it is at most $\min\{\bar{M}_i, D_i\}$. □

Proof of Proposition 3.2. Differentiating (14) gives $\Phi_i = x_i\big(x_i^2 - w_i(x_{-i})\big)$, which vanishes at $x = 0$ for every x_{-i}. Since $0 \in X_i = [-c, c]$, the projection fixes it, so $r_i(0) = \|\Pi_{X_i}(0)\| = 0$ for every player. Each $J_i(\cdot, x_{-i})$ is even in x_i with minimizers $\pm\sqrt{w_i(x_{-i})}$ and minimum value

$-\frac{1}{4}w_i(x_{-i})^2$, while $J_i(0, x_{-i}) = 0$. At the origin $w_i(0) = 1$, hence every player's regret is $\frac{1}{4}$. The condition on c places $\pm\sqrt{w_i}$ inside X_i for every $x_{-i} \in X_{-i}$. □

Proof of Proposition 4.1. Fix a player i, a covering point $z \in \mathcal{Z}$ and a coordinate k of player i's block, and write $\varphi(t) = J_i(z + te_{i,k})$, so that $\varphi'(0) = [\Phi_i(z)]_k$ is the quantity (15) estimates. The assumed extension makes φ three times continuously differentiable on $[-\tau, \tau]$, hence Taylor's theorem with Lagrange remainder gives $\varphi(\pm\tau) = \varphi(0) \pm \tau\varphi'(0) + \frac{1}{2}\tau^2\varphi''(0) \pm \frac{1}{6}\tau^3\varphi'''(\xi_\pm)$ for some $|\xi_\pm| \le \tau$. Subtracting the two expansions cancels the even orders, so that

$$\frac{\varphi(\tau) - \varphi(-\tau)}{2\tau} \;-\; \varphi'(0) \;=\; \frac{\tau^2}{12}\big(\varphi'''(\xi_+) + \varphi'''(\xi_-)\big). \tag{19}$$

Since $\varphi''' = \partial^3 J_i/\partial x_{i,k}^3$ is bounded by B_3 on the probe segment, the right side of (19) is at most $\frac{1}{6}B_3\tau^2$ in magnitude. Stacking the n_i coordinates, player i's bias vector has norm at most $\frac{1}{6}B_3\tau^2\sqrt{n_i}$.

For the noise, a difference of two independent σ-sub-Gaussian variables is $\sqrt{2}\sigma$-sub-Gaussian, so dividing by 2τ makes each difference quotient s-sub-Gaussian about its mean with $s = \sigma/(\sqrt{2}\tau)$. There are $d|\mathcal{Z}|$ scalar estimates in all, one per coordinate and covering point, and each has two-sided tail $2\exp(-t^2/2s^2)$. Equating $2d|\mathcal{Z}|\exp(-t^2/2s^2)$ with α and solving for t, a union bound gives that every coordinatewise deviation is at most $s\sqrt{2\log(2d|\mathcal{Z}|/\alpha)} = \frac{\sigma}{\tau}\sqrt{\log(2d|\mathcal{Z}|/\alpha)}$ simultaneously, with probability at least $1 - \alpha$; the $\sqrt{2}$ in s cancels the one from the tail. On that event player i's noise vector has norm at most $\frac{\sigma}{\tau}\sqrt{n_i\log(2d|\mathcal{Z}|/\alpha)}$.

Hence, on the same event and for every $z \in \mathcal{Z}$ at once, the triangle inequality bounds $\delta_i(z) = \|G_i(z) - \Phi_i(z)\|$ by the measured quantity $\|G_i(z) - \widehat{\Phi}_{\tau,i}(z)\|$ plus the bias and noise norms just obtained. Finally, given $x \in \Omega$ choose $z \in \mathcal{Z}$ with $\|x - z\| \le \eta$, which exists because $\mathcal{Z}$ is an η-covering of Ω; Lipschitzness of δ_i gives $\delta_i(x) \le \delta_i(z) + L_\delta\eta$. Taking the maximum of the measured term over the covering therefore yields (16), and the bound holds for every $x \in \Omega$ simultaneously. The argument for $\bar{M}_i$ is the same with M_i in place of δ_i, L_M in place of L_δ, and $\max_{z\in\mathcal{Z}}\|\widehat{\Phi}_{\tau,i}(z)\|$ as the measured term. □

D CONSTANTS AND RATES OF THE BLOCKWISE BOUND

Three analytic supplements to Lemma 3.1 and Theorem 3.1: what an average prediction error fails to control, whether the coefficients can be shrunk, and when the rate in $\rho_i + \delta_i$ improves from linear to quadratic.

D.1 AVERAGE PREDICTION ERROR DOES NOT CONTROL THE CONSISTENCY ERROR

Remark 2.1 asserts that a small training loss leaves the operator-consistency error of Lemma 3.2 uncontrolled. The following one-player construction makes the failure explicit and elementary, and shows that the obstruction is a gap between an average and a supremum rather than anything about multi-player coupling.

Proposition D.1 (Small mean-squared error, displaced fixed point)**.** *Let $X = [0, 1]$, let the reference response be $F_0 \equiv 0$, and for $0 < h < \frac{1}{4}$ let*

$$T_h(x) = \tfrac{1}{2} \max\big\{1 - |x - \tfrac{1}{2}|/h,\ 0\big\}. \tag{20}$$

Then under observations uniform on X, $\mathbb{E}\big|T_h(x) - F_0(x)\big|^2 = h/6 \to 0$ as $h \to 0$, while $\sup_{x \in X} \big|T_h(x) - F_0(x)\big| = \frac{1}{2}$ for every h. The map T_h retains the reference fixed point at 0 and acquires a second fixed point at $\frac{1}{2}$, whose fixed-point residual is zero and whose distance from the reference fixed point is $\frac{1}{2}$.

Proof. The triangle T_h is supported on $[\frac{1}{2} - h, \frac{1}{2} + h]$ with peak $\frac{1}{2}$ at $x = \frac{1}{2}$, so its supremum is $\frac{1}{2}$ and $\int_0^1 T_h^2 = h/6$ by direct integration of the squared triangular profile. Evaluating (20) gives $T_h(0) = 0$ and $T_h(\frac{1}{2}) = \frac{1}{2}$, hence both points are fixed. The residual at $\frac{1}{2}$ vanishes because $\frac{1}{2}$ is a fixed point. □

To read this as a statement about the consistency error of Lemma 3.2, instantiate the game: take $J(x) = \frac{1}{2}x^2$ on $X = [0, 1]$, hence $\Phi(x) = x$ and the reference response is $F_0 = \Pi_X(x - \Phi(x)) \equiv 0$. A head inducing $G(x) = x - T_h(x)$ gives $F = \Pi_X(x - G(x)) = T_h(x)$ and has $\delta = T_h$, so $\sup_X \delta = \frac{1}{2}$ while $\mathbb{E}\,\delta^2 = h/6$. In that game $M = D_1 = 1$, hence at $\hat{x} = \frac{1}{2}$ the certificate returns $(1 + 1)(0 + \frac{1}{2}) = 1$ against a model-free bound of 1: an exact tie, so that the residual entry buys nothing there by Corollary 3.1. The bound itself still holds, the true exploitability at $\hat{x} = \frac{1}{2}$ being $\frac{1}{8}$; what a small average error is compatible with is a displaced fixed point together with a residual entry that improves on nothing.

Three readings follow. An average prediction error of $h/6$ is compatible with a uniform map error of $\frac{1}{2}$, so that no bound on the former controls the latter without a uniform regularity hypothesis; the spurious fixed point at $\frac{1}{2}$ has zero residual, which is Remark B.4 in miniature and for a different reason; and the construction does not contradict any contraction result, since T_h has slope magnitude $1/(2h) > 1$ on its support. What is required to connect an average loss to the consistency error is a uniform statement, which is what the regional envelope of Proposition 4.1 supplies by probing a finite covering rather than by appealing to a training loss.

D.2 SHARPNESS OF THE BLOCKWISE CONSTANTS

Neither coefficient in (8) can be rescaled downward. One limitation bounds what follows and is worth stating before it rather than after: the family below has $\varepsilon = MD$ at every parameter value, so the model-free entry binds throughout and the residual entry never does. What is established is therefore sharpness of the coefficients at profiles where a practitioner would report the model-free bound instead.

Take $n = 1$, $X = [0, 1]$, $J(x) = ax$ and evaluate at $x = 1$, so that $M = a$, $D = 1$ and $r = \min\{1, a\}$. The attainment ratio $\varepsilon/[(M + D)r]$ equals $1/(1 + a)$ for $a \le 1$, which tends to one as $a \to 0$ and so forbids replacing D by cD with $c < 1$; and it equals $a/(1 + a)$ for $a \ge 1$, which tends to one as $a \to \infty$ and so forbids replacing M by cM. The argument excludes rescaling either coefficient and not a differently shaped constant.

D.3 THE QUADRATIC REGIME

Whether the rate in (12) is linear or quadratic turns on whether best responses lie in the relative interior of the action sets. Write $\mathrm{BR}_i(x_{-i}) = \arg\min_{y_i \in X_i} J_i(y_i, x_{-i})$.

Proposition D.2 (Quadratic certificate under interior best responses)**.** *Let Assum. 2.1 hold, let $J_i(\cdot, x_{-i})$ be differentiable with L_i-Lipschitz gradient $\Phi_i(\cdot, x_{-i})$ on X_i, and let some minimizer $b_i \in \mathrm{BR}_i(\hat{x}_{-i})$ lie in the relative interior of X_i for every player i. With $\rho_i = \rho_i(\hat{x})$ and $\delta_i^{\mathrm{BR}} \triangleq \|g_i(h_i(\hat{x}_{-i})) - b_i\|$,*

$$\varepsilon(\hat{x}) \;\le\; \max_{i \in \mathcal{N}} \tfrac{L_i}{2} \left(\rho_i + \delta_i^{\mathrm{BR}}\right)^2. \tag{21}$$

Proof. Fix a player i. Since $b_i \in X_i$ we have $\Pi_{X_i} b_i = b_i$, hence nonexpansiveness of Π_{X_i} gives $\|\hat{x}_i - b_i\| \le \rho_i + \delta_i^{\mathrm{BR}}$ as in Lemma 3.2. Minimality of b_i gives $\langle \Phi_i(b_i, \hat{x}_{-i}), y_i - b_i \rangle \ge 0$ for every $y_i \in X_i$. Because b_i lies in the relative interior of X_i, for each such y_i the point $b_i - t(y_i - b_i)$ also lies in X_i for all small $t > 0$, which supplies the reverse inequality; hence $\langle \Phi_i(b_i, \hat{x}_{-i}), y_i - b_i \rangle = 0$ for every $y_i \in X_i$, and in particular at $y_i = \hat{x}_i$. The descent lemma along the segment $[b_i, \hat{x}_i] \subseteq X_i$ then bounds player i's regret by $\frac{L_i}{2}\|\hat{x}_i - b_i\|^2$. Taking the maximum over players gives (21). □

Two scope statements belong with Proposition D.2. Relative interiority, rather than topological interiority, is the sharp hypothesis: at a relative-interior point of a lower-dimensional X_i the gradient need not vanish, but its inner product against every feasible direction does, which is all the proof uses. Playerwise convexity is not needed, hence (21) holds outside Assum. 2.2. What the refinement does require is a strictly stronger oracle than the envelope of Definition 3.1: δ_i^{BR} is a distance to an $\arg\min$, hence obtaining it means solving n constrained convex programs in the unknown costs. We therefore report (21) as the rate that governs the interior regime, not as a computable certificate, and Section 5 measures it only where ground truth is available.

E RESPONSE MODELS AND RECOVERING MULTIPLE EQUILIBRIA

Multiple equilibria and what recovers them. Non-monotone coupling makes multiple equilibria the common case even under Assum. 2.2: of 120 playerwise-convex instances, 113 carry at least two distinct Nash equilibria, verified by active-set enumeration to a worst residual of 1.42×10^{-14}. A linear regressor fitted to play pooled across both basins lands between them rather than at either, recovering both equilibria on a fraction 0.431 of instances. A unimodal nonlinear network recovers both on 0.862, against 0.224 for flow matching and 0.0517 for diffusion.

These numbers do not separate capacity from generativity, and we do not read them as doing so. Every family is evaluated through a single-valued map, which for the two generative families is a conditional mean rather than a sample, so the comparison is between mean maps and the question of set-valuedness is not posed. The parameter counts run the other way besides: the generative heads carry roughly four times the parameters per player of the unimodal network and still recover fewer equilibria. What the measurement does establish is narrower and still useful: a linear responder fitted across basins lands between them, and sufficient capacity avoids that failure. We record it because it severs any empirical benefit claim from the observation that set-valued responses sit inside the hypotheses of Theorem 3.1; the latter is a scope statement, not a performance one.

Set-valuedness where convexity fails. On the nonconvex witness of Proposition 3.2 the conditional-mean profile has exploitability 0.25 analytically while its variational residual is 0, so the residual alone certifies nothing there. Mode-selected samples reduce exploitability from 0.256 to 0.0286 for diffusion ($p_{\mathrm{Holm}} = 1.91 \times 10^{-6}$) and from 0.252 to 0.0175 for flow matching ($p_{\mathrm{Holm}} = 1.91 \times 10^{-6}$), over 20 paired seeds. The certificate's hypotheses fail on this game and we never evaluate the bound as a claim about it.

F EXPERIMENTAL SETUP

The studies named in Section 5 correspond to the modules of the released package as follows: the *tightness study* to `e1_tightness`, the *coupling sweep* to `e2_coupling`, the *dissociation study*

to `e3_incomparability`, the *set-valued witness* to `e4_set_valued`, the *multiplicity study* to `e5_multiplicity`, the *solver comparison* to `e6_solvers`, the *probe study* to `e7_probe`, the *scaling study* to `e8_scaling`, and the *amortization study* to `e9_amortization`.

Each game is chosen so that it isolates one hypothesis of Theorem 3.1 and supplies either analytic ground truth or a documented numerical oracle, so that a discrepancy is attributable to the bound rather than to an undocumented solver. We describe the five in turn, in each case stating what makes the instance hard and which claim it exercises.

Coupled quadratic: monotonicity on a knob. Set $Q_i = I$ and $A = I + \kappa(W + \frac{1}{2}Z)$ with W symmetric and Z antisymmetric, both with zero diagonal blocks, W rescaled so that $\lambda_{\min}(W) = -1$ and Z so that $\|Z\|_2 = 1$. Then

$$\tfrac{1}{2}\big(A + A^\top\big) = I + \kappa W, \qquad \text{hence} \qquad \lambda_{\min}\big(\tfrac{1}{2}(A + A^\top)\big) = 1 - \kappa, \tag{22}$$

so the monotonicity modulus is known analytically and the sweep crosses monotonicity exactly at $\kappa = 1$. The skew part keeps $A \neq A^\top$, so that the pseudo-gradient is not the gradient of any potential and the instance is not a potential game. This is the only family in which instances can be placed arbitrarily far on either side of the monotone boundary with the modulus known in closed form, which is what lets the coupling sweep attribute an outcome to non-monotonicity rather than to conditioning. It carries the claim that the certificate requires no monotonicity.

Capacitated Cournot: an active projection. Firms choose scalar quantities on $[0, \mathrm{cap}_i]$ with capacities alternating between 0.6 and 5.0, so that the tight capacities bind at equilibrium. The projection in r_i is therefore active, which is exactly the regime Proposition 3.1 excludes by hypothesis. The game is monotone and unremarkable as economics; its role is to show that the interiority hypothesis in Proposition 3.1 is load-bearing rather than decorative, since on this game the learned map *can* sharpen a pointwise certificate.

Rosen shared constraint: outside the product assumption. Boxes intersected with a single coupled row $w^\top x \leq h$, so that player i optimizes over $\mathcal{X}_i(x_{-i}) = \{x_i \in [\mathrm{lo}_i, \mathrm{hi}_i] : w_i^\top x_i \leq h - \sum_{j \neq i} w_j^\top x_j\}$. This violates the product requirement of Assum. 2.1, so every result here is formally silent about it. We include it to record what the certificate does when its own hypothesis fails, and we report those cells as a stress diagnostic and never as coverage. Two consequences are worth naming: the best response is a projection onto a box intersected with a halfspace, which is computed by exact bisection on the single multiplier and so needs no inner solver; and D_i is conservative here, since it is the diameter of the box rather than of the coupled set.

Bimodal coordination: convexity removed. With J_i and w_i as in (14), $\partial^2 J_i / \partial x_i^2(0) = -w_i < 0$, so playerwise convexity fails. All 2^n sign patterns of $s = (1 - \kappa)^{-1/2}$ are equilibria in closed form, while the origin is a stationary point of Φ at which the variational residual vanishes and the exploitability is exactly $\frac{1}{4}$. This is the witness of Proposition 3.2. It tests the *necessity* of Assum. 2.2 and not the validity of the bound, and we never evaluate (12) as a claim about it.

Nonexpansive boundary: monotone yet not contractive. Let $J_i(x) = \frac{1}{2}(x_i - \sigma_i x_{i+1})^2$ cyclically with $\sigma_i = +1$ for $i < N - 1$ and $\sigma_{N-1} = -1$. Then $F(x) = Px$ with P a signed cyclic permutation, so that $P^\top P = I$ and the best-response modulus is exactly one: nonexpansive, and not a contraction. The equilibrium is nevertheless unique and interior at the origin, because $P^N = (\prod_i \sigma_i) I = -I$ makes $I - P$ invertible. Picard iteration is $2N$-periodic and therefore never converges from any $x^0 \neq 0$, its residual staying exactly constant. This is the sharpest separation we can construct between monotonicity of the variational inequality and convergence of naive fixed-point iteration: the game is strongly monotone with a unique equilibrium, and the iteration still fails. It is the regime in which contraction-based guarantees say nothing while Theorem 3.1 still applies, and it is where the solver comparison is run.

Ground truth and evaluation protocol. Every game supplies a closed-form or verified-global best response, so that no inner minimization is approximated and the exploitability against which each bound is checked is exact rather than a solver output. Equilibria are verified to a residual below 10^{-10} when an equilibrium is required: the capacitated Cournot equilibrium by active-set enumeration, the

coupled-constraint variational equilibrium by projected extragradient to its documented tolerance, and the bimodal game's 2^n equilibria in closed form.

The nine experiments use the following fixed protocol. E1 tests bound validity and tightness on the three games above (420 cells, of which 280 satisfy the theorem assumptions); E2 sweeps eight coupling values on the product quadratic game (1,120 cells); E3 pools only the five families with rigorous Lipschitz bounds to compare joint and small-gain contraction certificates; E4 tests the nonconvex set-valued witness; E5 enumerates equilibria and compares responder families; E6 compares Picard, Krasnosel'ski–Mann, and Halpern on the analytic nonexpansive map; E7 varies finite-difference radius and oracle noise; E8 varies player count at a fixed observational-data budget; and E9 varies covering resolution and region dimension. Unless stated otherwise, E1–E2 use seeds 0–19, seven response families, 200 training epochs, and 240 observational transitions per seed. E3–E9 use the seed counts, tolerances, held-out profile counts, and correction procedures reported with their respective results files. All comparisons pair the same seed, game, family, and coupling wherever those factors are present; confidence intervals use $10{,}000$ paired bootstrap resamples, and Wilcoxon tests use Holm correction within the declared experiment family.

Response families. Seven response models receive the concatenated rival profile x_{-i} and are fitted on observational play under a shared budget, so that no family is advantaged by a longer fit: a least-squares affine map, two-layer 32–32 tanh and ReLU networks, a spectral-normalized network, an Almost-Orthogonal-Layer (AOL) Lipschitz-certified network (Prach & Lampert, 2022), a conditional diffusion sampler, and conditional flow matching. The first five expose a rigorous Lipschitz bound computed from weights; the two samplers expose none, so that their certificate status is unknown and is recorded as not-certified rather than as passing. Capacity is not matched across the two groups. The deterministic heads use two hidden layers of width 32 and the generative ones two of width 64 together with a time embedding, so that the generative arm carries roughly four times the parameters per player — a mismatch that runs against, rather than for, any reading in which capacity explains the outcomes of Appendix E. All neural responders use Adam with batch size 128 for 200 epochs on the same 240 noisy observational transitions per seed; the learning rate is 3×10^{-3} except for AOL, which uses 5×10^{-3}. Affine and feedforward responders minimize mean-squared prediction error, diffusion uses denoising-score matching, and flow matching uses its conditional flow objective.

Table 1: **Certificate divided by true exploitability as players are added.** Coupled quadratic game, scalar actions, medians over 20 seeds; lower is tighter and one is exact. The aggregated column is the single-constant form $(M + D)(\rho + \delta)$, the blockwise column is Theorem 3.1. The aggregated form grows with n because exploitability is a maximum over players whereas an aggregated residual is a joint norm. In this game the blockwise column coincides with the model-free bound $\max_i \bar{M}_i D_i$, hence it reports tightness of the model-free entry, not of the residual entry.

players n	2	8	64	128	256
aggregated	38.15	84.85	431.1	696.1	1157
blockwise	13.92	10.78	7.862	7.176	6.628

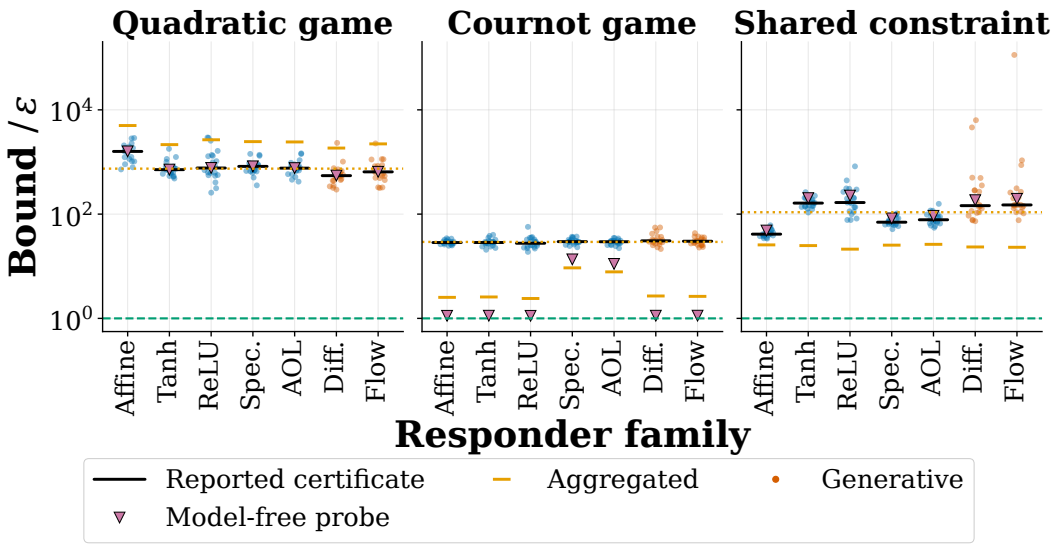


Fig. 4: **Tightness by response family.** Certified bound over true exploitability, dots seeds and bars medians, log scale; panels are the quadratic, Cournot and shared-constraint games. Blue marks families with a rigorous Lipschitz bound, orange those without.

G Robustness and Scope

Five figures, grouped evidence are reported here to support our main results. Fig. 4 and Fig. 5 test whether the pointwise result survives a change of architecture and game parameter. Fig. 6 calibrates the estimator supplying one term of the envelope. Fig. 7 and Fig. 8 bound the scope: the first exhibits a hypothesis failing, the second a familiar sufficient condition.

G.1 Invariance across response families and coupling

Fig. 4 plots the certified bound of (12) over the true exploitability, one dot per seed and one bar per median, for seven response families on three games. The families sit close together: the widest within-game spread is a factor of 4.04. The reported certificate equals its model-free entry $\max_i \bar{M}_i D_i$ in all 280 product-set cells and in the 140 shared-constraint diagnostic cells; the latter are not theorem coverage. The measured fraction is 1. The plotted ratio therefore carries no learned-map contribution; families differ only through $\bar{M}_i$, which records where each fixed point lands.

Fig. 5 sweeps the coupling parameter, which fixes $\lambda_{\min}$ of the symmetrized pseudo-gradient Jacobian exactly, so that the grid crosses monotonicity at one. Validity is invariant: the bound holds in 1120 of 1120 cells across the grid, the non-monotone half included, which is the content of requiring no monotonicity. Tightness is not: its median falls monotonically from $3.01e+03$ to 9.86. The sweep establishes robustness of validity and nothing about the quality of the bound. The bound holds at every coupling; the median tightness ratio, on the right axis, falls monotonically by a factor of about 300 across the grid, so what the sweep shows to be stable is validity and not tightness. Certified fractions are shown for the joint-spectral and small-gain conditions.

G.2 Calibration of the central-difference estimator

Fig. 6 reports the central-difference error in estimating Φ_i against the probe radius τ at five noise scales, with medians over twelve probe points at each radius and, at each nonzero scale, twenty-four noise replicates. Panel (a) has $B_3 > 0$ and exhibits the interior minimum that the bias–noise balance of (16) predicts; panel (b) sets $B_3 = 0$ exactly, where no interior minimum can arise and none does. Dotted verticals mark the balance radius that Proposition 4.1 predicts, one per noise scale. The observed minimizer over that radius is 2.14, against a grid of multiplicative step 2.15. The predicted

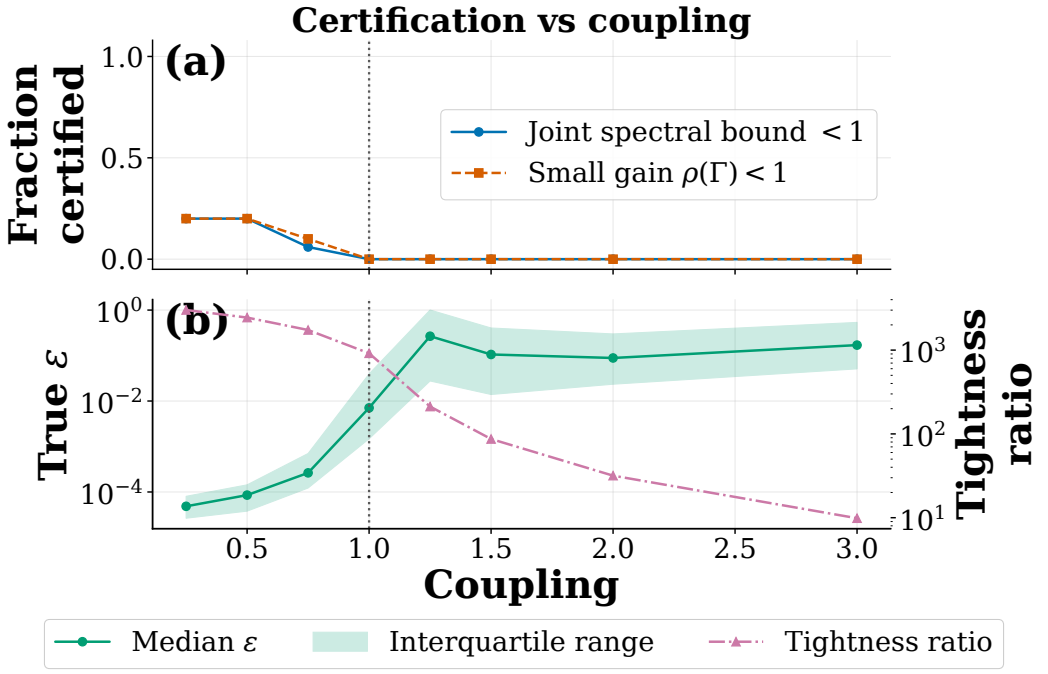


Fig. 5: **Coupling sweep across the monotone boundary.** Coupled quadratic game; the coupling parameter sets $\lambda_{\min}$ of the symmetrized pseudo-gradient Jacobian exactly, so the grid crosses monotonicity at one.

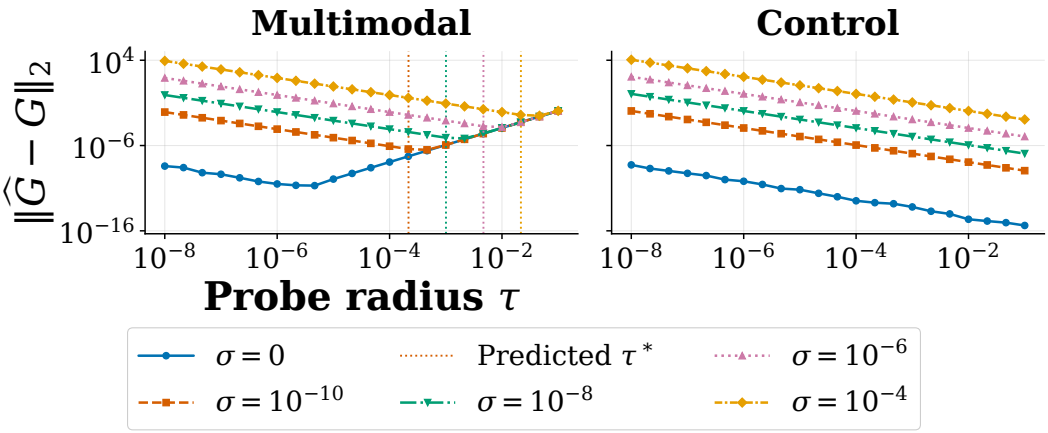


Fig. 6: **Probe error against probe radius.** Central-difference error in estimating Φ_i versus τ, at several noise scales σ; medians over twelve probe points and twenty-four noise replicates each.

scaling in σ is confirmed to within one grid step; the prefactor is not resolved, the grid being coarser than the discrepancy under test. Panel (a) is the multimodal arena, where $B_3 > 0$ and the predicted U-shape appears: bias of order $B_3\tau^2$ dominates at large τ and noise of order σ/τ at small, with the minimum near the balance point of Proposition 4.1. Panel (b) is a control with $B_3 = 0$ exactly, in which no interior minimum can occur and none does. The curves are diagnostics of the estimator, not certificates.

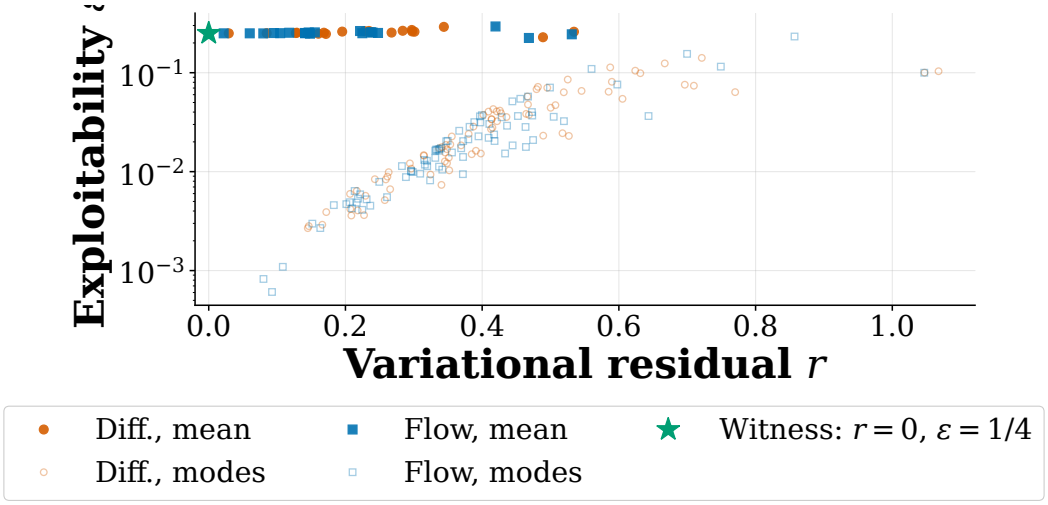


Fig. 7: **A vanishing residual is no evidence of a small gap.** Exploitability against variational residual on the bimodal witness of Proposition 3.2, log ordinate, 20 seeds per family. Filled markers are the conditional-mean fixed points, hollow markers the four mode-selected profiles per seed, and the star is the analytic witness, at residual exactly zero and exploitability $1/4$. Mode selection moves the cloud down and to the right of the conditional-mean profiles it improves on, so that a smaller gap is bought at a larger residual. Within the mode-selected cloud the two quantities are positively associated; it is the shift between the groups that the plane shows.

G.3 Scope: Convexity failure and an incomparable condition

Fig. 7 compares the exploitability reached from a conditional-mean response with that reached from mode-selected samples, on the bimodal witness of Proposition 3.2, over 20 paired seeds.The structure of interest is at the origin, where the variational residual vanishes analytically while the exploitability is 0.25. The learned conditional-mean fixed points land beside it, at 0.256 and 0.252, with measured residual 0.1834 rather than the analytic zero, so that the degeneracy is approached and not attained. Mode selection lowers exploitability to 0.0286 and 0.0175 without repairing the hypothesis: recovery of the game's 4 equilibria is bimodal across runs, with more than half recovering none, so that no run is typical. Fig. 7 places all of this in the residual–exploitability plane, and the ordering there is the point. The witness sits at residual zero with exploitability 0.25, while the mode-selected profiles reach those smaller gaps carrying a median residual of 0.3721, larger than the 0.1834 of the conditional-mean profiles they improve on. The comparison is paired and between groups rather than monotone: a fraction 0.825 of mode-selected profiles carry both a smaller exploitability and a larger residual than the conditional-mean profile of the same seed, so that a smaller gap is bought at a larger residual. Within the mode-selected cloud the residual and the gap remain positively associated, so that the residual does not even order the profiles it fails to control.

Fig. 8 cross-classifies the joint-spectral against the small-gain condition over 1100 certifiable cells pooled from the two sweeps: 106 certified by both, 4 by small gain alone, 0 by the joint condition alone, and 990 by neither. The empty cell is a fact about these fits rather than a theorem, since $\rho(\Gamma) \leq \|J_F\|_2$ held in all 1100 of them with median slack 0.03969, so that the direction separating the two conditions analytically is realized nowhere in this population. All 106 doubly certified cells are affine. Reporting the joint-spectral condition alone would understate coverage on fits of this kind, which is not a dominance claim: both separating directions are realizable, and the two analytic witnesses accompanying the contingency distinguish an empty cell from an impossible one.

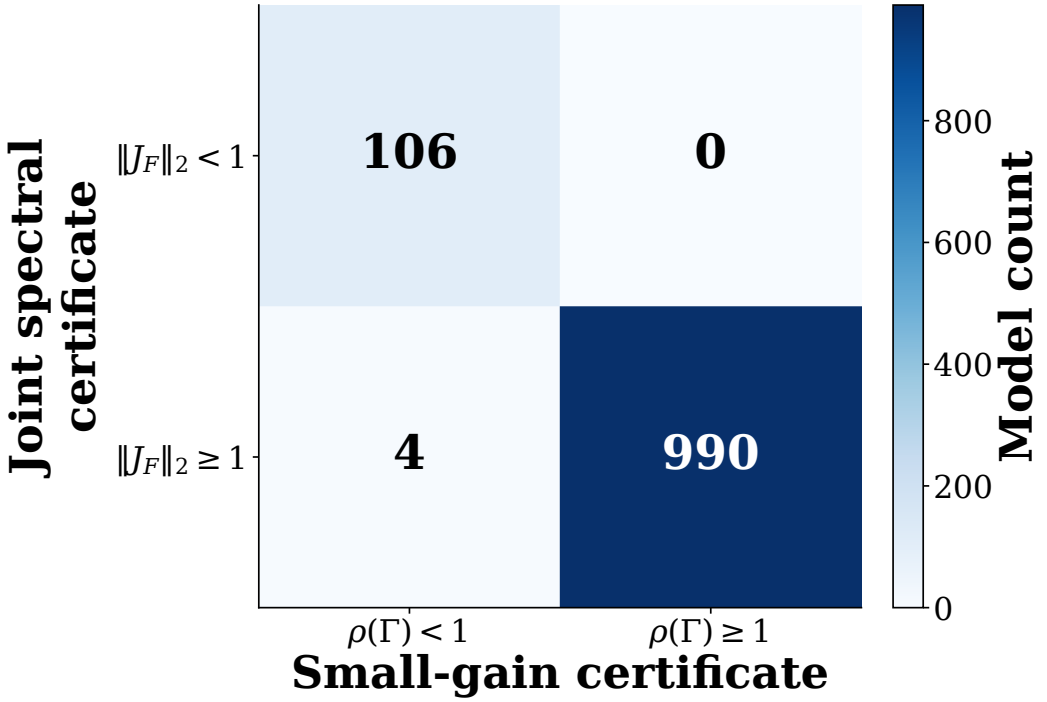


Fig. 8: **Joint-spectral against small-gain certification on trained models.** Contingency counts pooled over games and seeds, over 1100 certifiable trained models: 106 certified by both, 4 by small gain alone, 0 by the joint bound alone, and 990 by neither. The empty cell is a fact about these fits and not a dominance relation: $\rho(\Gamma) \leq \|J_F\|_2$ held in all 1100 of them, with a median gap of 0.040, so the skew-coupling direction that separates the two conditions analytically does not arise here at any threshold. This licenses no claim that either certificate dominates.